

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Research Agent

Domain of Project – Education

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Problem Statement - Research Agent

The Challenge

Researchers and students spend significant time searching for relevant papers, summarizing findings, managing citations, and identifying research gaps. With the growing volume of academic literature, finding accurate and reliable information has become increasingly difficult.

The Objective

Build an **AI-powered Research Agent** that assists users in conducting efficient academic and scientific research. The solution should:

Literature Search – Retrieve relevant research papers and articles.

Paper Summarization – Generate concise summaries of research documents.

Citation Management – Create and organize references in standard formats.

Research Gap Analysis – Identify limitations and future research opportunities.

Hypothesis Generation – Suggest innovative research ideas.

Report Generation – Generate literature reviews and research reports.

Technology Used

IBM Watsonx Orchestrate – Used to design and manage the multi-agent research workflow, coordinating different research tasks efficiently.

IBM Granite Model (Granite 3.x) – Used for natural language understanding, research paper summarization, hypothesis generation, and report writing.

IBM Cloud – Provides a secure, reliable, and scalable platform for deploying and managing the Research Agent.

Retrieval-Augmented Generation (RAG) – Retrieves relevant information from research papers and knowledge sources before generating accurate responses.

IBM Watsonx.ai – Used for model deployment, prompt engineering, embeddings, and AI-powered research assistance.

Technology Stack Summary

IBM Watsonx Orchestrate

IBM Granite 3.x

IBM Watsonx.ai

IBM Cloud

RAG (Retrieval-Augmented Generation)

Proposed Solution - AI-Powered Research Agent

The proposed solution is a **multi-agent AI-powered Research Agent** built using **IBM Watsonx Orchestrate**, **IBM Granite Models**, **IBM Watsonx.ai**, and **RAG** to simplify academic and scientific research. The system helps users search research papers, summarize findings, manage citations, identify research gaps, and generate research reports efficiently.

The solution uses a **RAG-based knowledge retrieval system** to fetch relevant information from research papers and academic sources before generating responses. IBM Granite models are used for reasoning, summarization, and content generation.

The system consists of five key agents:

Literature Search Agent – Retrieves relevant research papers and articles.

Research Summarization Agent – Generates concise paper summaries.

Citation Management Agent – Creates and organizes references.

Research Gap Analysis Agent – Identifies limitations and future opportunities.

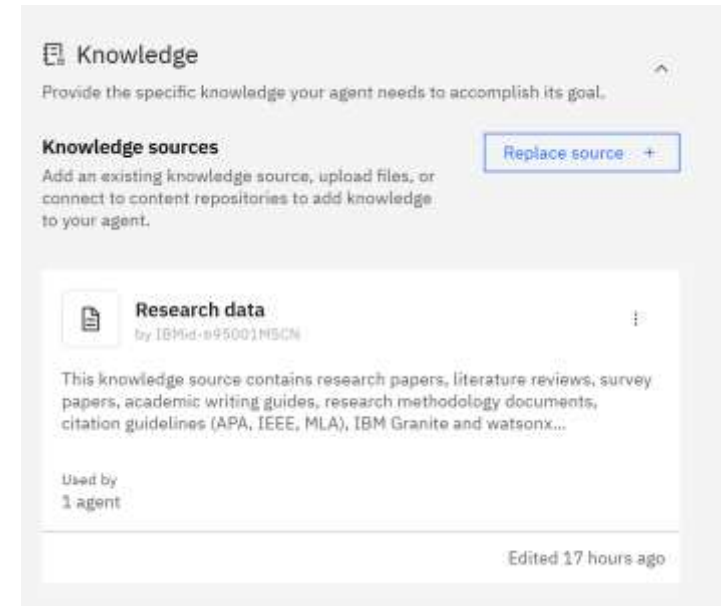
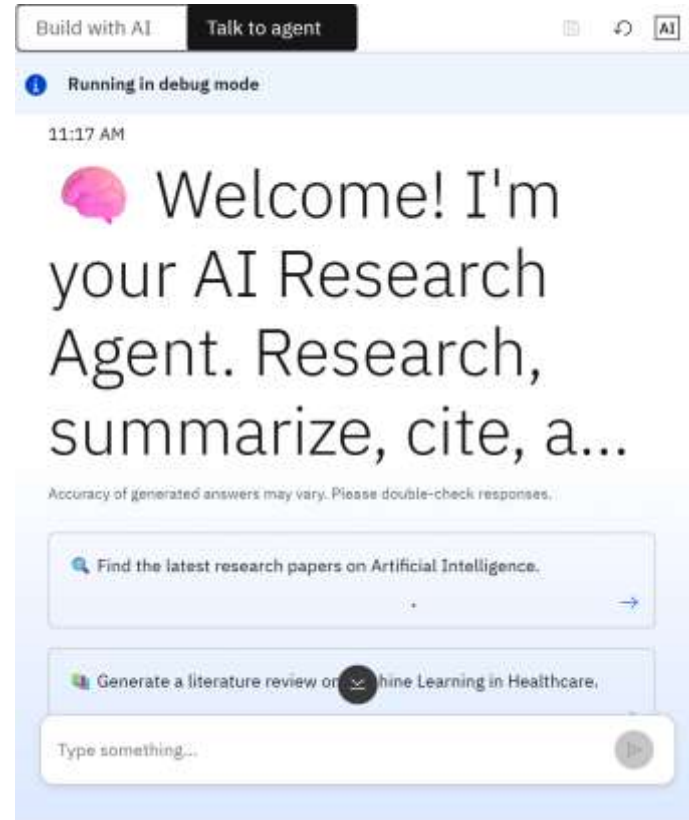
Report Generation Agent – Produces literature reviews and research reports.

IBM Watsonx Orchestrate manages agent workflows, while **IBM Watsonx.ai** provides AI services and embeddings. Overall, the solution delivers an intelligent research assistant that improves research efficiency, accuracy, and innovation for students, researchers, and professionals. ✍️📝

Watsonx Orchestrate Components Used

- 1) Chat Input** – Interface where users enter research questions, paper topics, keywords, or upload research documents.
- 2) IBM Watsonx AI Agent** – Processes user queries using IBM Granite models and generates research insights, summaries, and recommendations.
- 3) Name of IBM Granite Model – Granite-3.3-8B-Instruct** → Used for research paper summarization, reasoning, citation generation, and report creation.
- 4) Chat Output** – Displays AI-generated responses such as paper summaries, literature reviews, research gaps, citations, and reports.
- 5) File Component** – Stores and retrieves research papers, journals, academic articles, and knowledge-base documents for RAG-based retrieval.
- 6) Knowledge Base / RAG Component** – Retrieves relevant information from uploaded research documents and trusted academic sources before generating responses.

- Screenshots –
- Research Agent Created & Knowledge Base



Agent Preview & Deployment :

You 11:36 AM

Generate a literature review on Agentic AI

Research Agent 11:36 AM | Show Reasoning ▾

Literature Review – Agentic AI (Autonomous, Reasoning-Enabled Language-Model Agents)

Prepared for: [User-requested] – concise yet comprehensive academic overview

1. Introduction

Agentic AI refers to **autonomous software agents that combine large-language-model (LLM) reasoning with concrete actions in an environment** (e.g., web browsing, API calls, robot control). Recent work demonstrates that coupling *verbal* reasoning traces with *act* primitives yields more transparent, robust, and flexible decision-making than treating LLMs as pure “black-box policies”. This review synthesizes the state-of-the-art, categorises methodological families, highlights empirical findings, and

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AICTE Agentic AI Challenge 2026

AI Research Agent

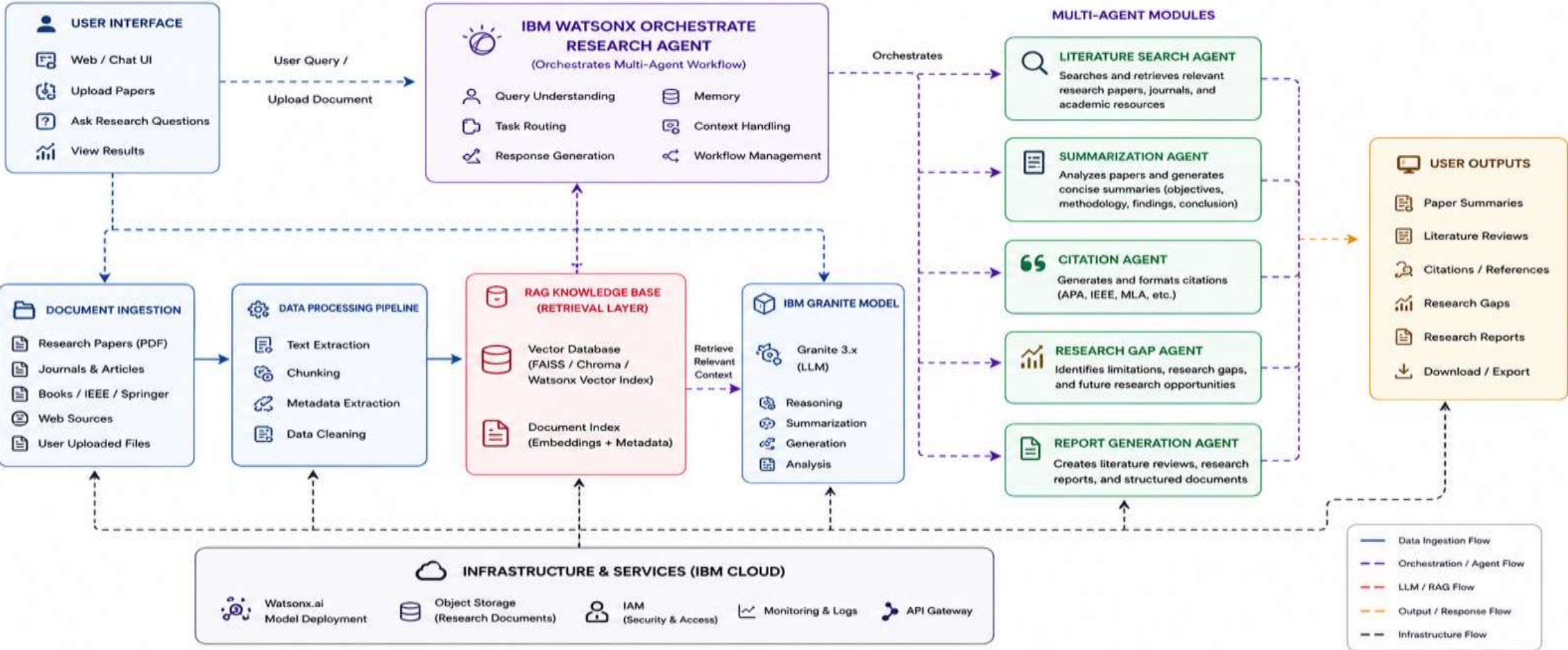
An intelligent research assistant powered by IBM Watsonx Orchestrate and Granite Models. Search literature, summarize papers, identify research gaps, generate citations, draft reports, suggest hypotheses, and accelerate scientific discovery.

Explore Features

 Literature Search  Summarization  Citations  Research Gaps  Hypotheses

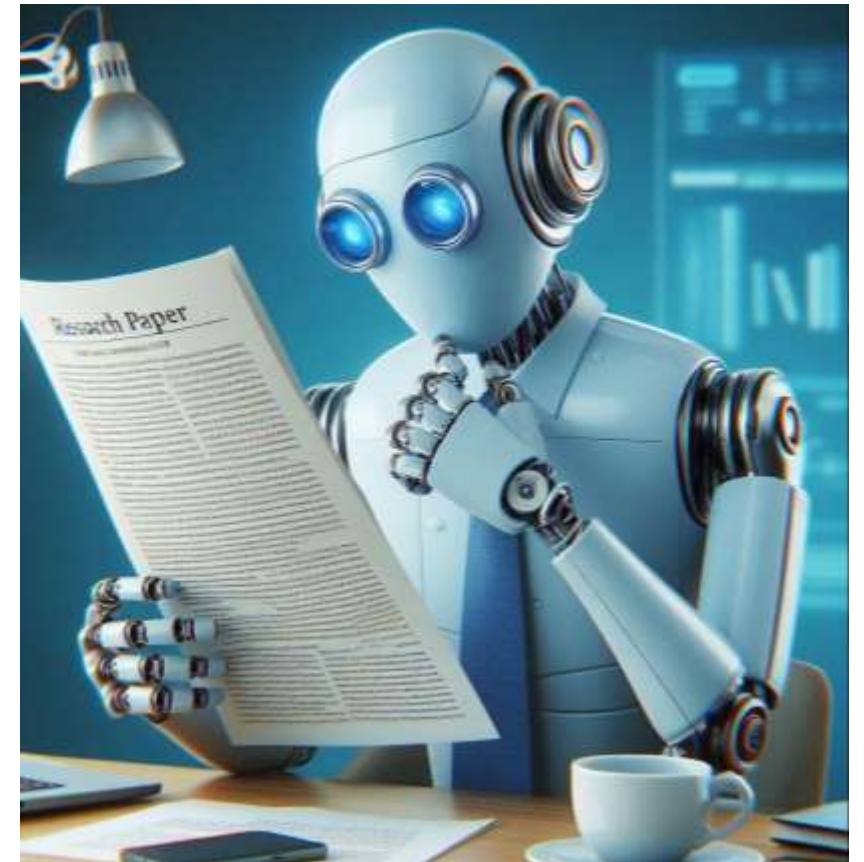
Architecture blueprint

Architecture Blueprint – Research Agent



Role of Agentic AI in Research Agent

- Agentic AI enables the Research Agent to function as an intelligent, autonomous system rather than a simple chatbot. Using IBM Watsonx.ai and IBM Granite Models, it coordinates multiple specialized agents to perform research-related tasks efficiently.
- Each agent handles a specific responsibility such as literature search, paper summarization, citation generation, research gap analysis, and report creation while collaborating with other agents to deliver accurate results.
- This multi-agent approach improves task execution, enhances reasoning, and reduces the time required for academic and scientific research. Agentic AI also provides context awareness by understanding user queries, research domains, and document content to generate relevant responses.
- It supports real-time adaptation by continuously refining outputs based on user feedback and newly available information. Through collaboration between agents, the system can make better decisions and provide more comprehensive research insights.
- Overall, Agentic AI transforms the Research Agent into a proactive and goal-driven research assistant that accelerates knowledge discovery, improves productivity, and supports evidence-based innovation.



Novelty and Uniqueness

The proposed **Research Agent** stands out by combining **Agentic AI, Retrieval-Augmented Generation (RAG), and IBM Granite Models** into a single intelligent research platform. Unlike traditional search tools, the system not only retrieves information but also understands, analyzes, and generates meaningful research insights.

Multi-Agent Intelligence – Multiple specialized agents collaborate to perform literature search, paper summarization, citation generation, research gap analysis, and report creation.

RAG-Based Accuracy – Integrates real-time knowledge retrieval from research papers, journals, and uploaded documents before generating responses, ensuring reliable and evidence-based outputs.

Research Gap Identification – Helps researchers discover limitations, unexplored areas, and future research opportunities from existing literature.

Automated Research Assistance – Reduces manual effort by automating time-consuming tasks such as literature review, citation management, and report generation.

Scalable & Enterprise-Ready – Built on IBM Watsonx Orchestrate, IBM Watsonx.ai, and IBM Granite Models, ensuring security, scalability, and reliability.

This makes the solution a **smart, autonomous, and efficient research assistant** that accelerates knowledge discovery and improves research productivity.

Git Hub Link

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Github URL – <https://github.com/ranaveer53/AI-Research-Agent.git>

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Future Scope

1. Integration with Academic Databases

The Research Agent can be integrated with academic databases such as IEEE Xplore, Springer, ScienceDirect, and Google Scholar to access the latest research publications and improve literature retrieval.

2. AI-Powered Voice Assistant & Multilingual Support

Future versions can support voice-based interactions and multiple languages, allowing researchers worldwide to access research assistance more conveniently and efficiently.

3. Advanced Research Gap Detection

The system can be enhanced to automatically identify emerging trends, unexplored research areas, and potential opportunities for innovation.

4. Collaborative Research Workspace

Researchers can collaborate, share documents, generate joint reports, and track research progress within a unified platform.

5. Automated Paper Writing Assistance

The agent can assist in drafting research papers, generating abstracts, formatting citations, and preparing publication-ready documents.

Thank You!

Thank you for your time and interest.